



HR Leave Management System

Group 2: Aya, Moamen, Gihad

The Cost of Manual Leave Management

Payroll Errors

Incorrect balances → costly mistakes

Compliance Gaps

No audit trail → legal risk

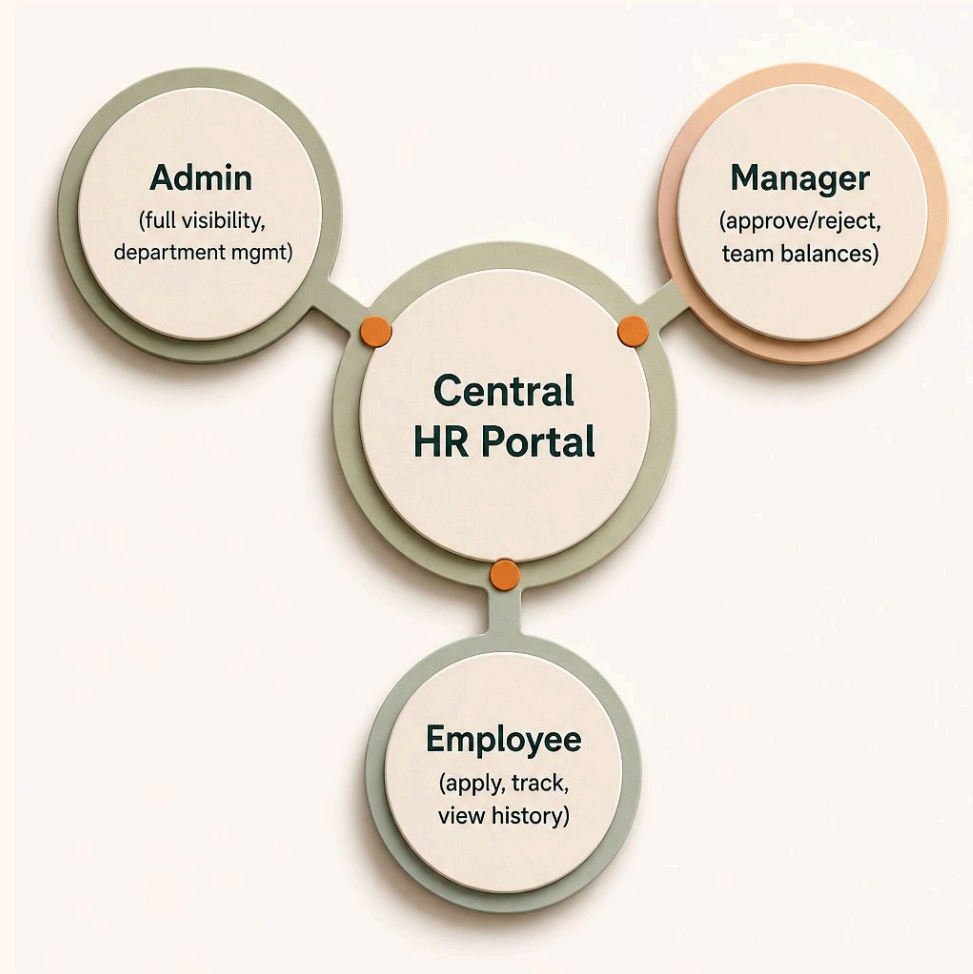
HR Bottlenecks

Manual reconciliation → wasted hours



One Portal. Three Roles. Real-Time.

Replaces spreadsheets and email chains with a single, role-aware HR portal.



Who Uses It



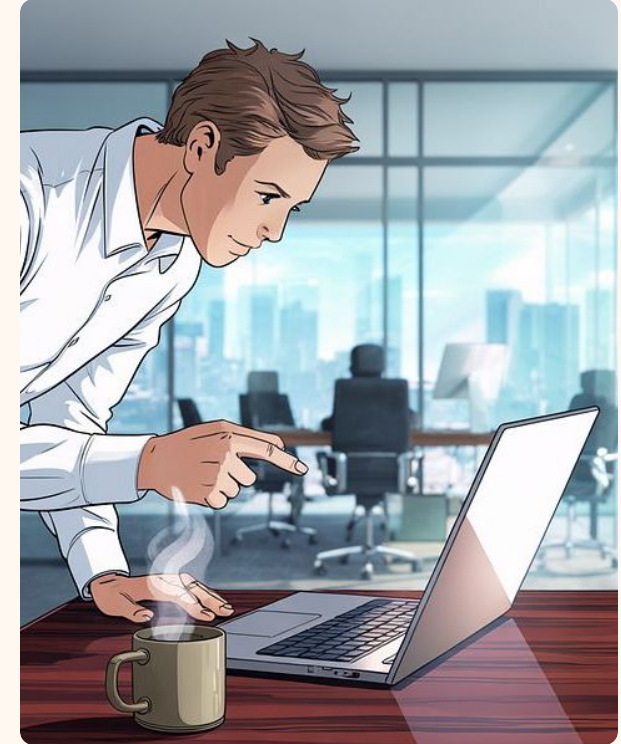
Admin

Live dashboards, department management, all leave requests



Manager

Approve/reject with comments, team balances



Employee

Apply for leave, track status, view balances

Admin Dashboard — Live at a Glance



6 KPI Cards

Employees, Departments, Requests,
Pending, Approved, Rejected

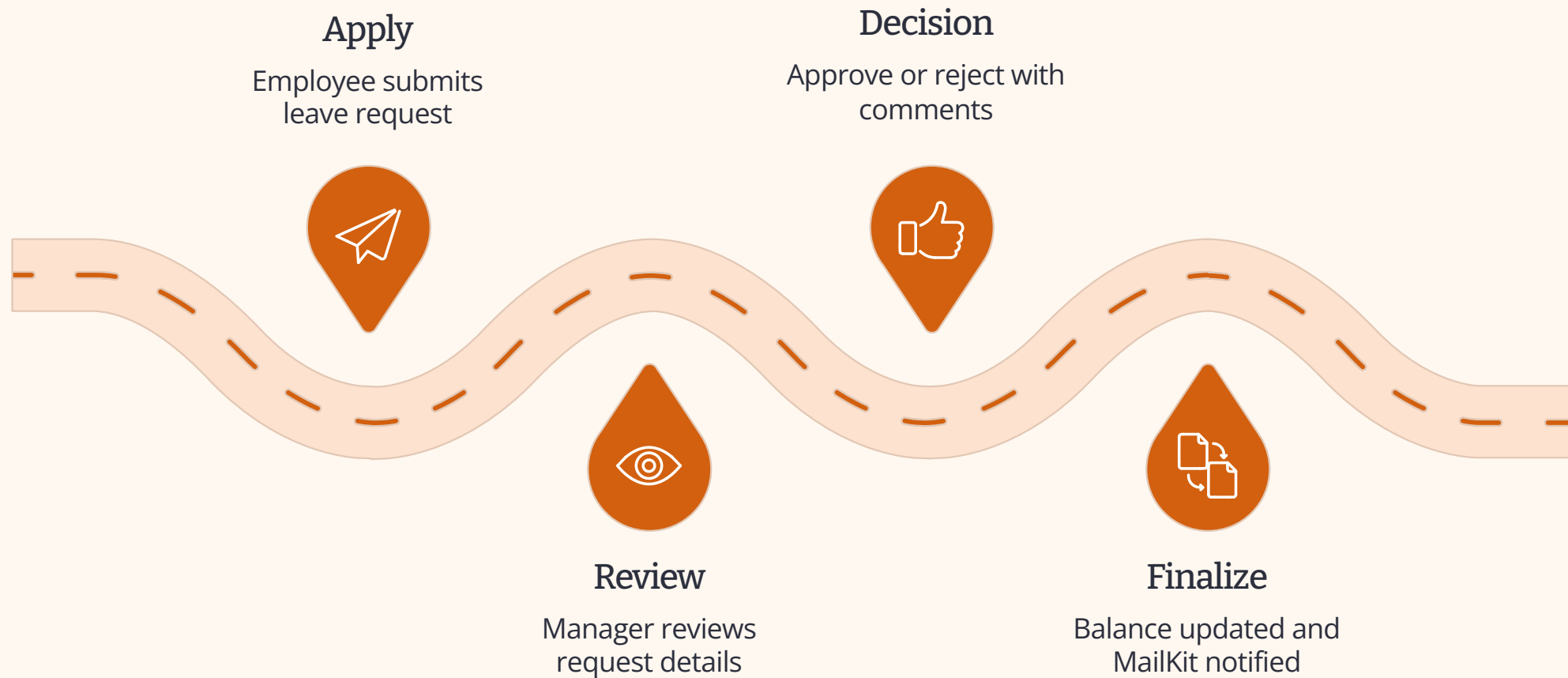
Department Usage Chart

Leave days by department —
Recharts bar chart

Leave Type Breakdown

Most-used leave types — Recharts pie
chart

Leave Lifecycle — End to End



Zero manual reconciliation. Every status change triggers an automated MailKit email to the relevant party.

Key Features



Role-Based Access

ASP.NET Identity enforced at every API level



Leave Balances

Real-time updates, no manual entry



Email Alerts

MailKit notifications on every status change



Departments

Full department management & reporting



Leave Types

Configurable types per organization



Public Holidays

Nager.Date API — zero manual entry

Technology Stack

Frontend

- React 19 · TypeScript · Vite 7
- TailwindCSS v4 · TanStack Query v5
- Recharts · Radix UI / shadcn
- React Hook Form · Zod · Framer Motion

Backend

- ASP.NET Core (.NET 10) · EF Core 10
- SQL Server · ASP.NET Identity
- JWT Bearer · BCrypt.Net · AutoMapper
- MailKit · Serilog · Swagger

Security & Data Architecture



Security Architecture

- JWT Bearer + refresh token rotation
- BCrypt.Net — industry-standard hashing
- Role-guard middleware on every endpoint
- User Secrets to hide all secrets

Data Model — EF Core 10 + SQL Server

- ApplicationUser, Department, LeaveType, LeaveRequest, EmployeeLeaveBalance, Holiday, RefreshToken.

Live Demo

Pack Your Seatbelt && Keep your head in the game